Lesson for the course “Human-Computer Interaction”

Visual perception (Part 1)

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The importance of cognitive psychology in IxD practice

- Interacting with technology involves a number of **cognitive processes**
  - Attention, perception, recognition, memory, reading, speaking and listening, problem-solving, planning, reasoning, decision-making, learning
- The knowledge of cognitive theories, modelling tools, and methods can lead to the design of better interactive products
  - We need to take into account **characteristics and limitations of users**
  - Predict what users can and cannot be expected to do
  - Identify and explain the nature and causes of problems users encounter

Our perception is biased

- Our perception of the world around us is not a true depiction of what is actually there
- Our perceptions are heavily biased by at least three factors
  - The past: **our experience**
  - The present: **the current context**
  - The future: **our goals**

- What do you see here?



- Do you see the building map?



- Do you see the word?



Perception biased by experience

- **Perceptual priming**



FIGURE 1.1 Building map or word? What you see depends on what you were told to see.

- When your perceptual system has been **primed** to see building shapes, you see building shapes, and the white areas between the buildings barely register in your perception
- When your perceptual system has been primed to see text, you see text, and the black areas between the letters barely register

Perception biased by experience

- **Familiar perceptual patterns**
  - Perceptual patterns are formed from repeated exposure to situations
  - …so our mind **expects** to find that pattern in presence of that situation
  - Because of the perceptual patterns users of computer software and websites have, they often click buttons or links **without looking carefully at them**; their perception of the display is based more on what their frame for the situation leads them to expect rather than on what is actually on the screen
  - This sometimes confounds software designers, who expect users to see what is on the screen – but that isn't how human vision works

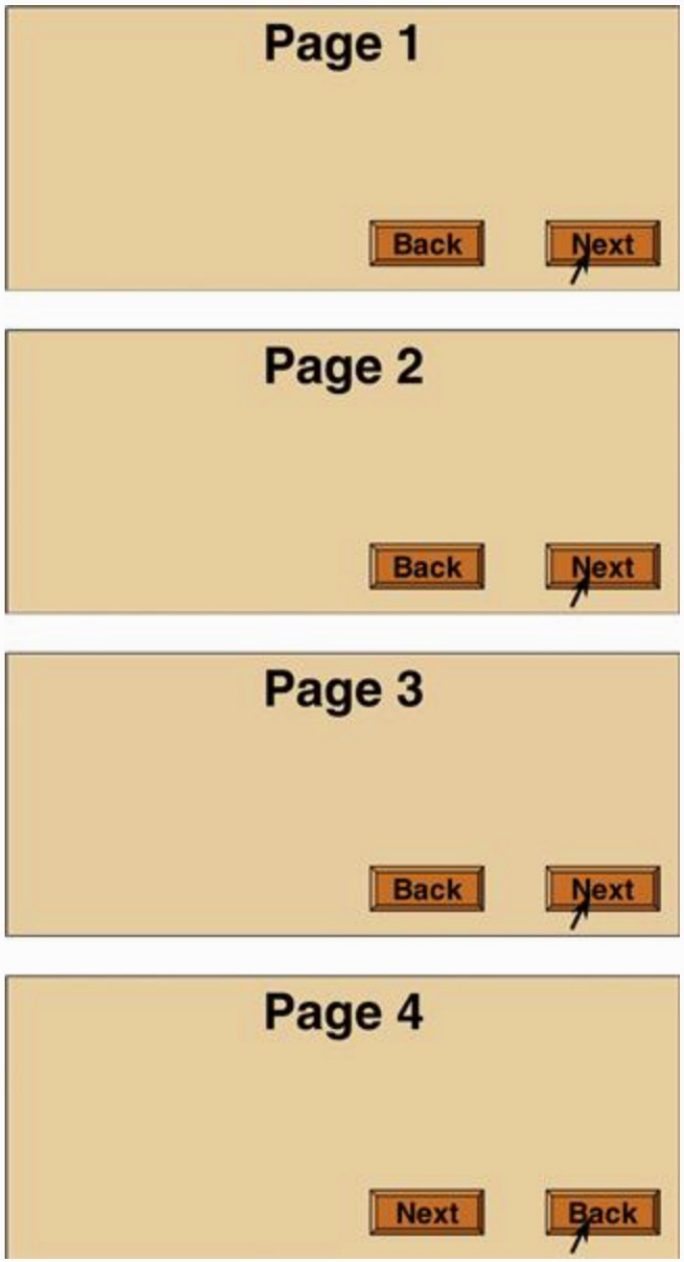


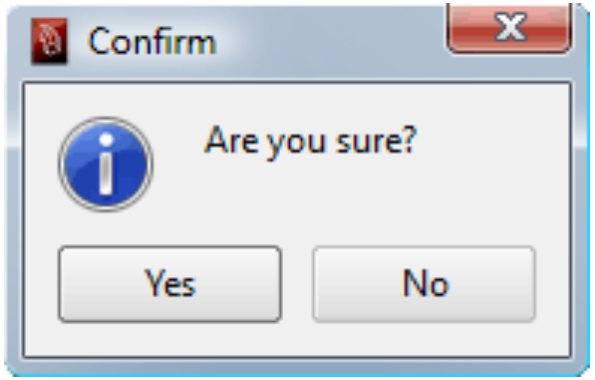
FIGURE 1.3 The "Next" button is perceived to be in a consistent location, even when it isn't.

- Guideline: **consistent placement** of controls ensures that reality matches the user's perceptual pattern for the situation



Perception biased by experience

- **Habituation**
  - Repeated exposure to the same (or highly similar) perceptions dulls our perceptual system's sensitivity to them



- E.g., we experience habituation in computer usage when the same error messages or "Are you sure?" confirmation messages appear again and again. People initially notice them and perhaps respond, but eventually click them closed reflexively without bothering to read them….

Perception biased by current context

- Reading letters can be biased by the meaning of a sentence

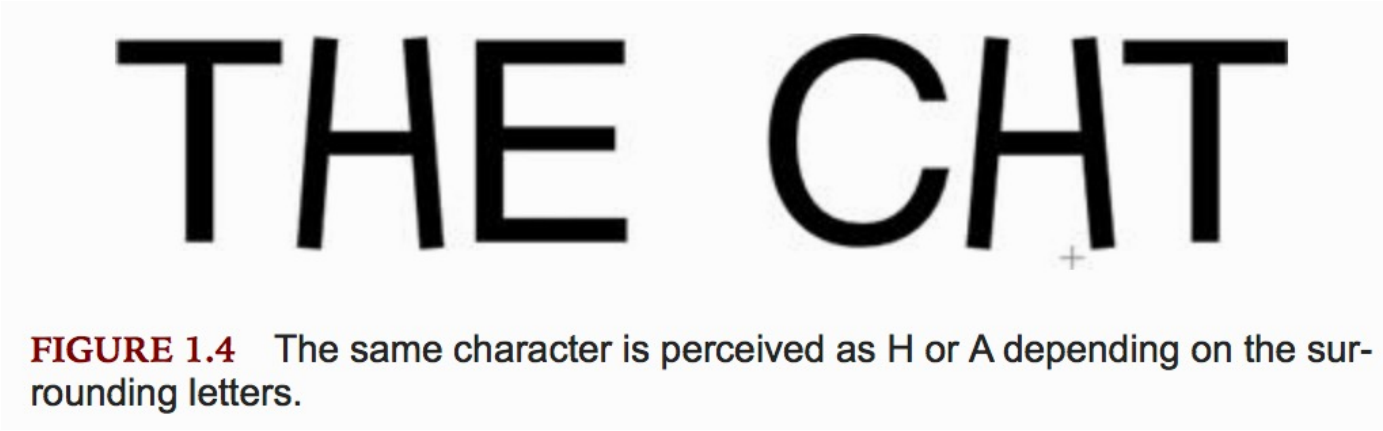


FIGURE 1.4 The same character is perceived as H or A depending on the surrounding letters.

Perception biased by current context

- Illusion can occur when vision is biased by context

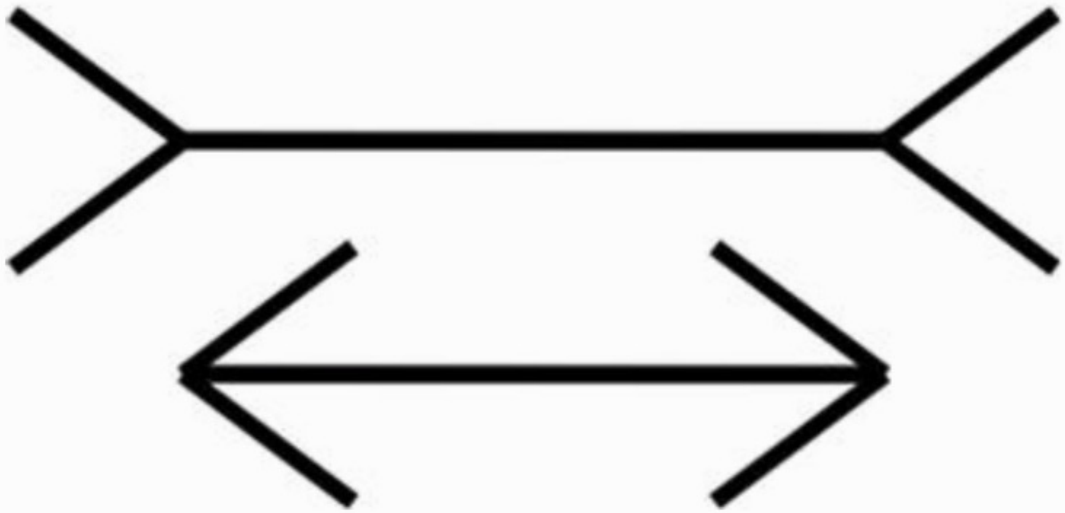


FIGURE 1.6 Müller–Lyer illusion: equal-length horizontal lines appear to have different lengths.

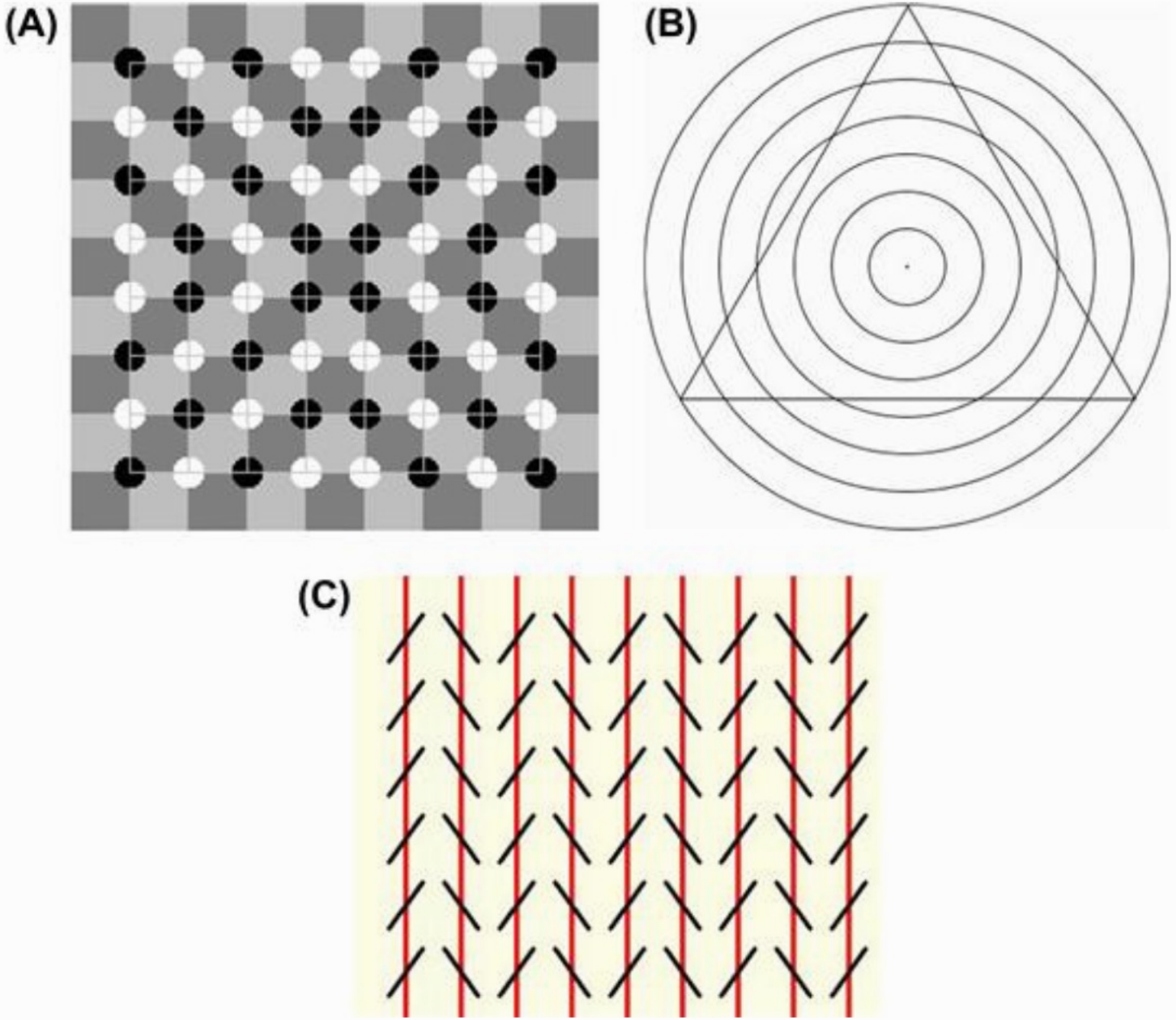


FIGURE 1.7 (A) The checkboard does *not* bulge in the middle; (B) the triangle sides are *not* bent; and (C) the red vertical lines are parallel.

Perception biased by goals

- Our goals:
  - **Guide our perceptual apparatus**, so we sample what we need from the world around us
  - **Filter our perceptions**: things unrelated to our goals tend to be filtered out preconsciously, never registering in our conscious minds

Perception biased by goals: example

- When people navigate through a Web site, seeking information or a specific function, they don't read carefully every word
  - They scan screens quickly and superficially, and consider the items that seem related to their goal
  - They don't simply ignore items unrelated to their goals; they often don't even notice them

Perception biased by goals: examples

- "Cocktail party effect": if you are conversing with someone at a crowded party, you can focus your attention to hear mainly what he or she is saying even though many other people are talking near you
  - The more interested you are in the conversation, the more strongly your brain filters out surrounding chatter
  - If you are bored by what your conversational partner is saying, you will probably hear much more of the conversations around you



Implications of perceptual biases for user interface design

- **Avoid ambiguity**
  - Avoid ambiguous information displays, and test your design to verify that all users interpret the display in the same way
  - Where ambiguity is unavoidable, either rely on standards or conventions to resolve it, or prime users to resolve the ambiguity in the intended way

Implications of perceptual biases for user interface design

- **Be consistent**
  - Place information and controls in consistent locations
    - Controls and data displays that serve the same function on different pages should be placed in the same position on each page on which they appear
    - They should also have the same color, text fonts, shading, and so on
  - This consistency allows users to spot and recognize them quickly



e.g., menu tabs consistent in different pages of a website

Implications of perceptual biases for user interface design

- **Understand the goals of the users**
  - Users come to a system with goals they want to achieve. Designers should understand those goals
  - Realize that users' goals may vary, and that their goals strongly influence what they perceive
  - Ensure that at every point in an interaction, the information users need is available, prominent, and maps clearly to a possible user goal, so users will notice and use the information

Gestalt psychology for visual interfaces design

- The Gestalt principles describe how the **visual system** groups elements
- Gestalt Principles can be used to
  - Impose a logical structure on user interfaces
  - Help people create logical groupings and increase visibility and comprehensibility of user interfaces
- We will see how to use spaces in visual design with Gestalt Principles
  - E.g. web design, software UIs

Our vision is optimized to see structure: the Gestalt principles

- The Gestalt principles describe how the **visual system** groups elements
  - **Figure/Ground**
  - **Proximity**
  - **Similarity**
  - **Focal point**
  - **Continuity**
  - **Closure**
  - **Symmetry**
  - **Common fate**

Figure-ground

- Our mind separates the visual field into the figure (the foreground) and ground (the background)
- The foreground consists of the elements of a scene that are the object of our primary attention, and the background is everything else



FIGURE 2.16 Figure/Ground: when objects overlap, we see the smaller as the figure and the larger as the ground.

Figure-ground

- **Pop up:** the old content that was the focus of attention, becomes temporarily the background



FIGURE 2.19 Figure/Ground is used at PBS.org's mobile Web site to pop up a call-to-action "over" the page content.

Figure-ground

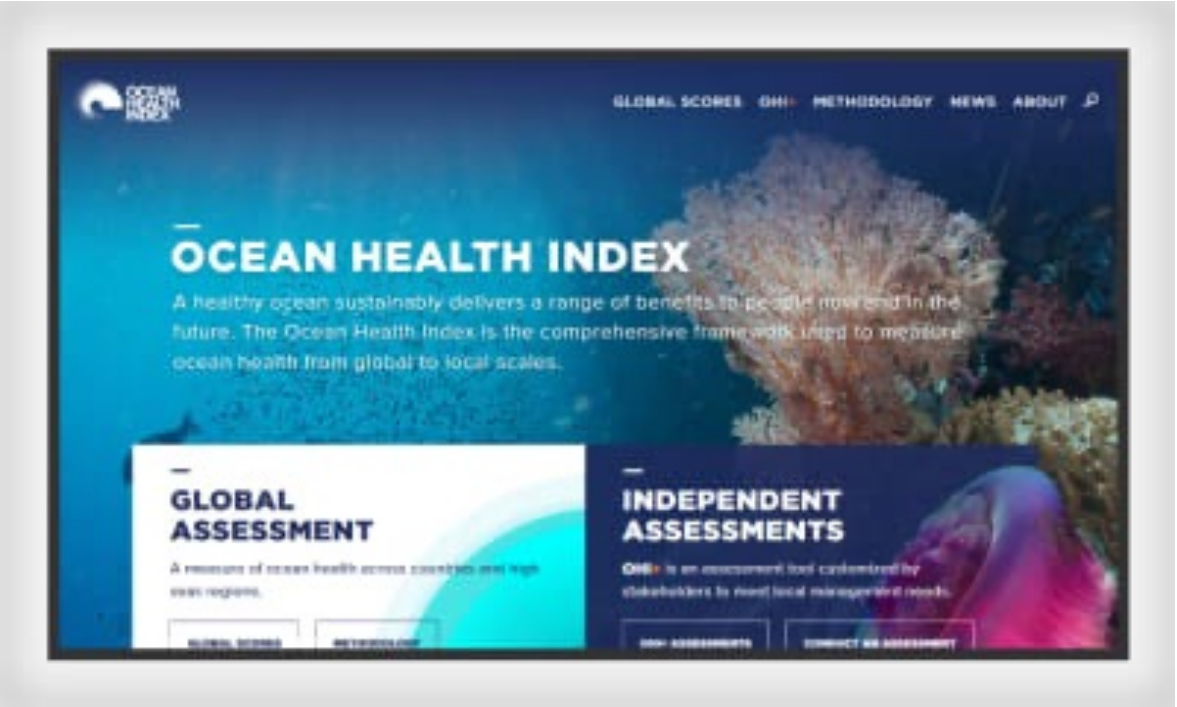
- The text is the figure in a large cursive font **in the center** of the page, making it **stand out** as compared to the image in the background. The white text over the contrasting grayscale image draws the focus of attention, which makes the text visible over it. The background is **blurred** so that the text pops out as the figure.



- All of these aspects show that **the text here is prioritized as the figure**, while the blurred image is used as background

Figure-ground

- This website makes a smart use of boxed content to separate the figure from the background

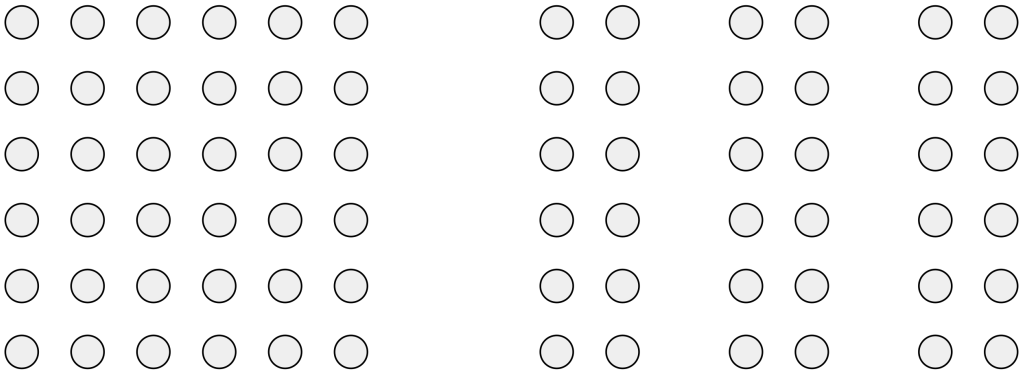


- The boxes down below are part of the figure, due to the contrast they offer to the ground image.

- The other aspect of this figure is **the second figure-ground contrast** that is placed inside the boxes in the form of contrastive-colored text.

Proximity

- Objects placed close together tend to be perceived as a group



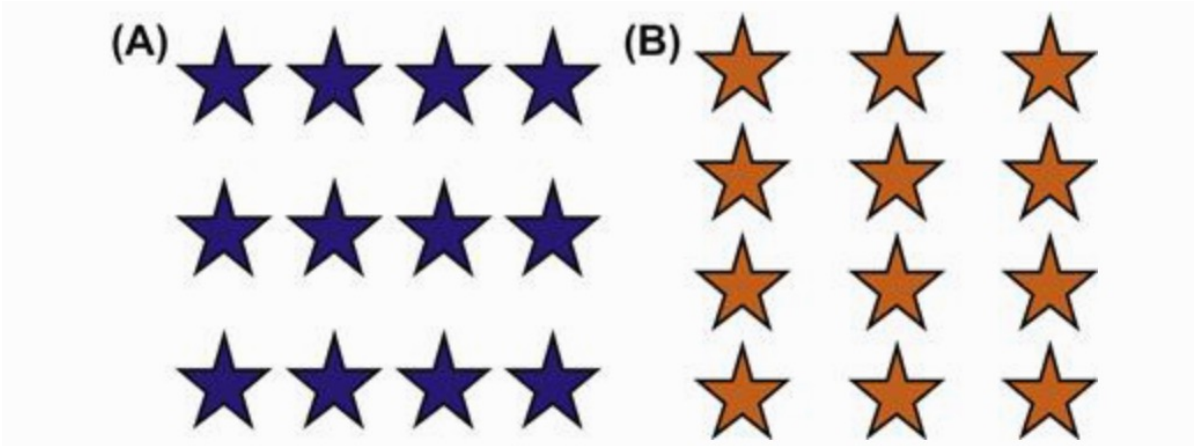
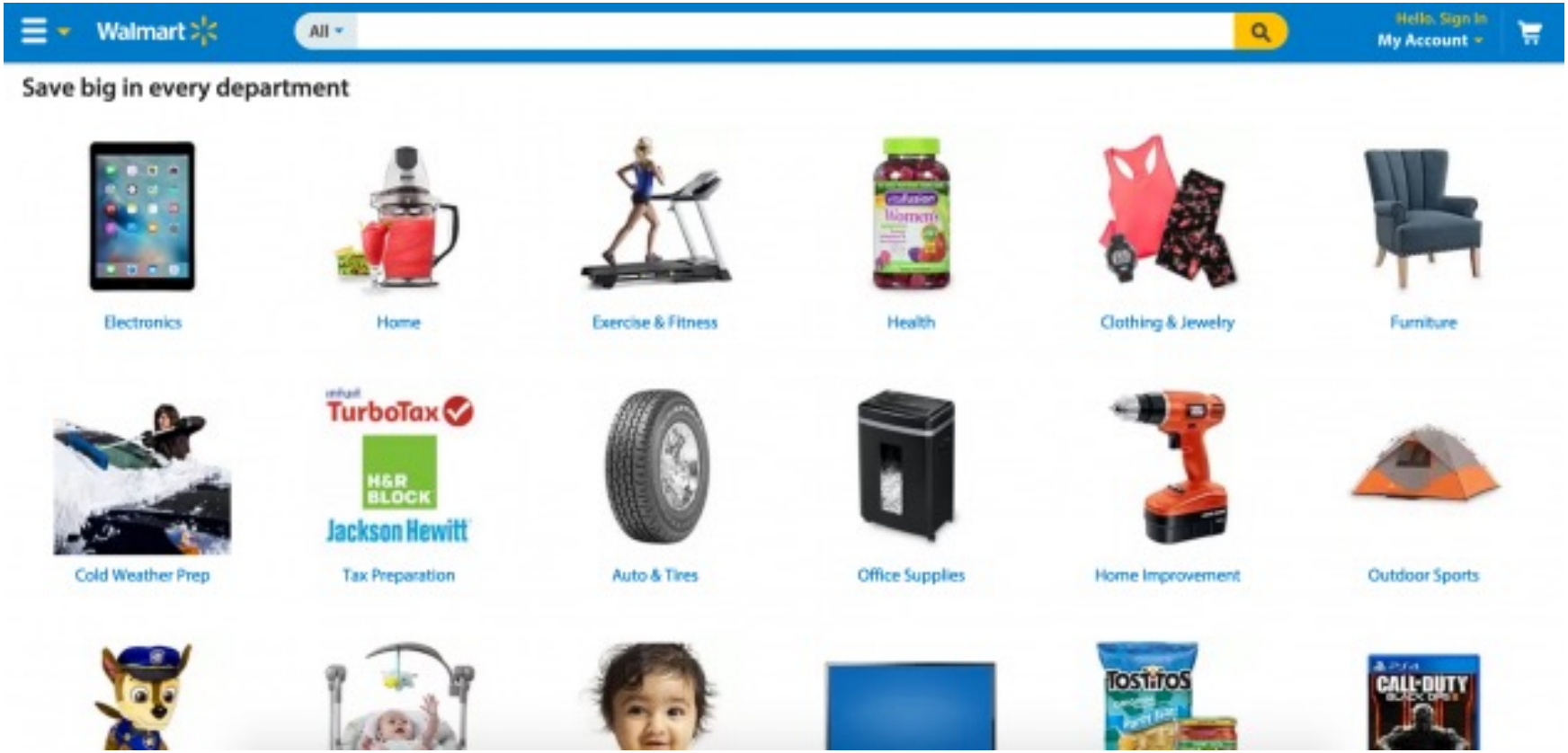


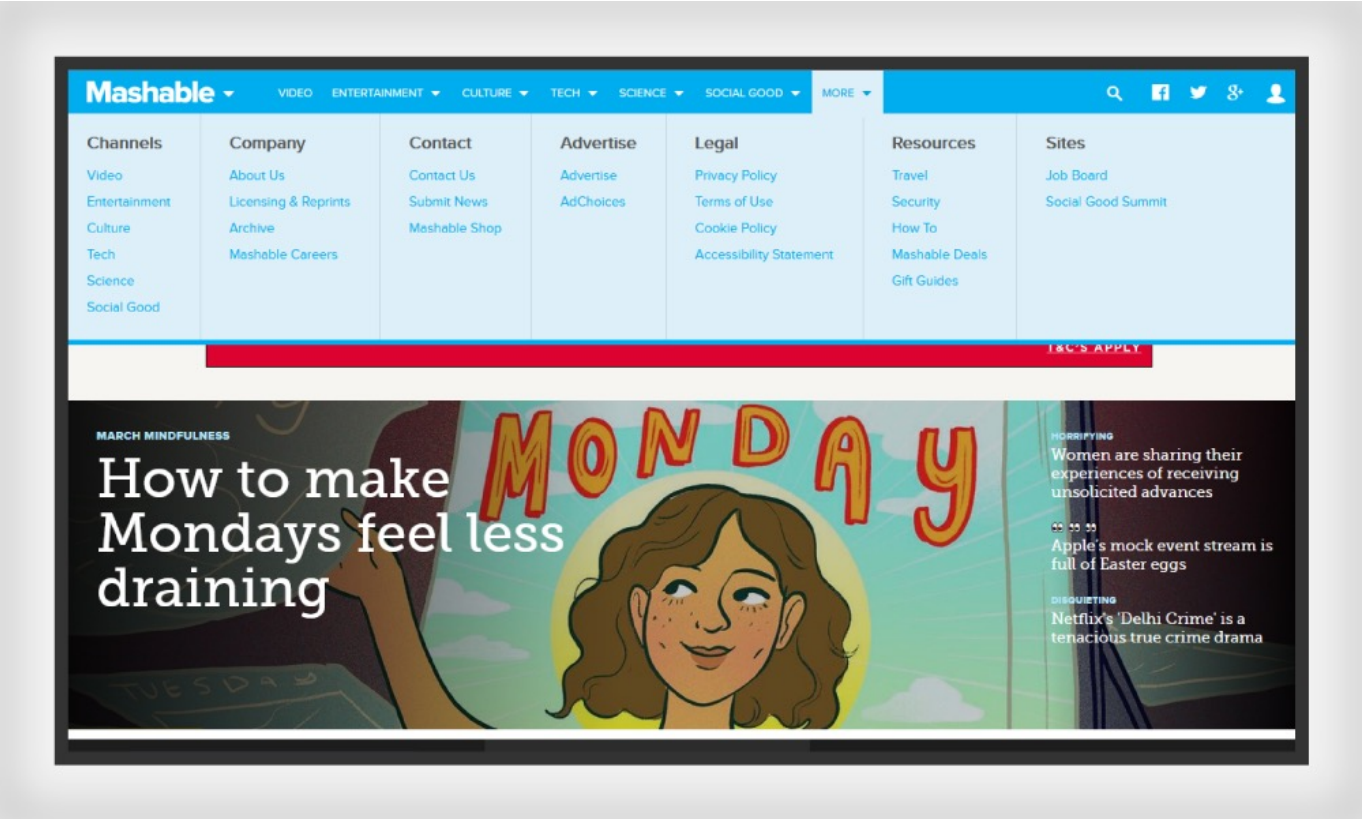
FIGURE 2.1 Proximity: items that are closer appear grouped as rows (A) and columns (B).

Proximity

- The nearness of each image and its corresponding text communicates that they're related to one another



Proximity



- You can clearly notice how the elements of the same category are placed together to show the relation with the main menu type. The users' eyes are automatically drawn to the subtypes under the main heading — also **thanks to the use of colors and text size.** You can see that the items that are unrelated are set apart with the help of a line, which serves as the users' guide in relating and separating the items as per the principle of proximity.

Proximity

- By grouping these images from the Apple Store together in close proximity, we assign a relationship to the whole group. We perceive them as being related and expect the content to be similar.



Proximity

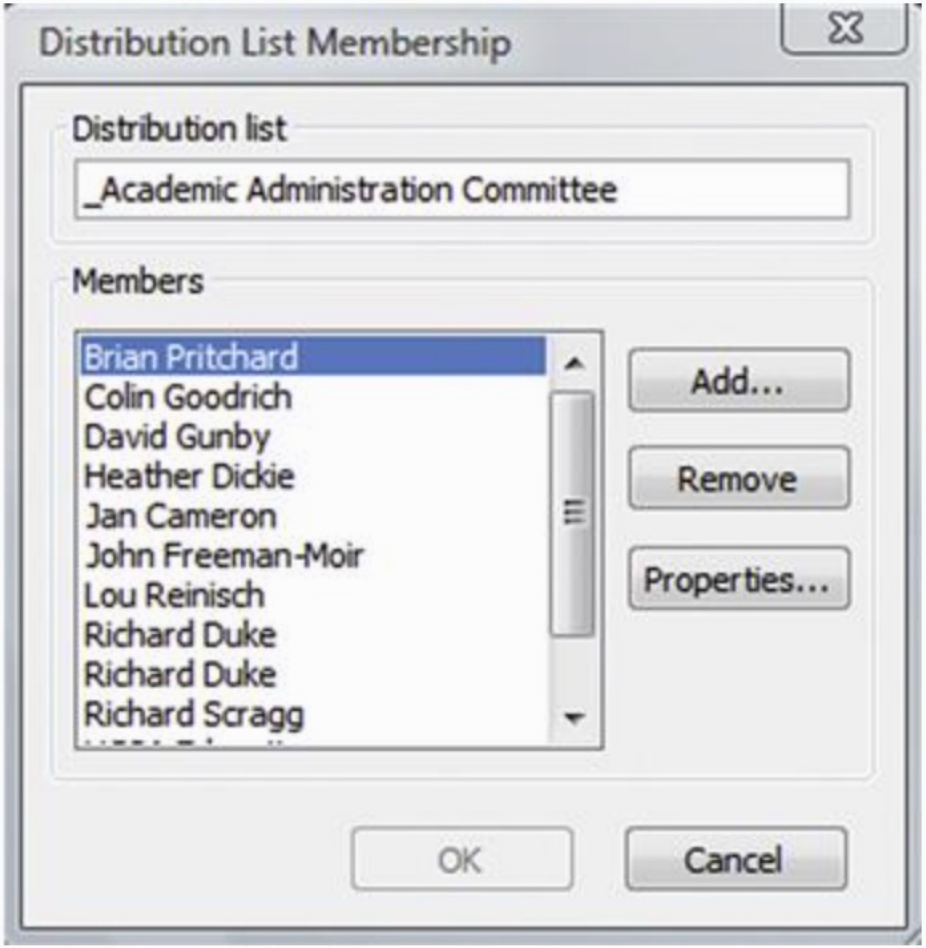


FIGURE 2.2 In Outlook's Distribution List Membership dialog box, list buttons are in a group box, separate from the control buttons.

Designers often separate groups of on-screen controls and data displays by enclosing them in group boxes or by placing separator lines between groups

Proximity



FIGURE 2.3 In Mozilla Thunderbird's Subscribe Folders dialog box, controls are grouped using the Proximity principle.

- However, according to the Proximity principle, items on a display can be visually grouped simply by spacing them closer to each other than to other controls, without group boxes or visible borders
- Many graphic design experts recommend this approach to reduce visual clutter and code size in a user interface

Proximity

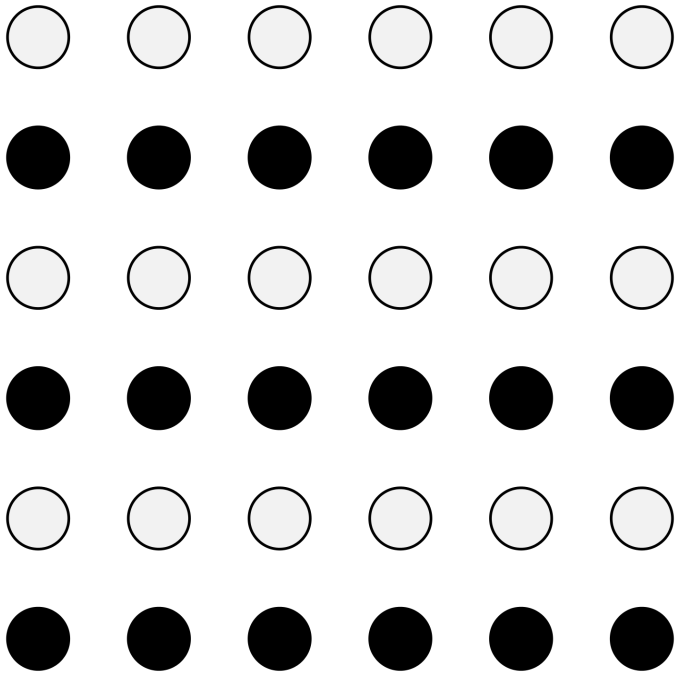


FIGURE 2.4 In Discreet's Software Installer, poorly spaced radio buttons look grouped in vertical columns.

- If controls are poorly spaced (e.g., if connected controls are too far apart) people will have trouble perceiving them as related, making the software harder to learn and remember
- Example: six horizontal pairs of radio buttons, each representing a two-way choice, but their spacing, due to the Proximity principle, makes them appear to be two vertical sets of radio buttons, each representing a six-way choice

Similarity

- Objects that look similar appear grouped, all other things being equal



Similarity

SALON NEWS POLITICS ENTERTAINMENT LIFE TECH BUSINESS SUSTAINABILITY INNOVATION Sign in

TOP STORIES | MOST RECENT



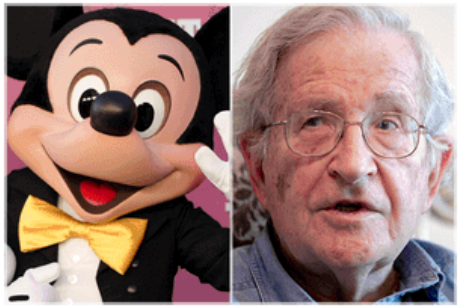
You cheat, you pay: How new contracts are changing love and religion forever

Love and contracts aren't strange bedfellows at all -- and sometimes it's the deals we make that keep us together

MARTHA M. ERTMAN

MONDAY, MAY 25, 2015 05:00 PM CDT

LIFE LOVE AND SEX, BOOKS, THE LAW



The right's big history lie: Reagan, Disney, Vietnam and the war to redefine America

The battle over history is the battle to define America's story. This Memorial Day, it is more important than ever

ANDREW HARTMAN

MONDAY, MAY 25, 2015 11:00 AM CDT

NEWS BOOKS, CULTURE WAR, RONALD REAGAN



The Republican campaign to destroy the poor stoops to a new low

The party of the rich is now doing everything in its power to make sure those without money live in misery

HEATHER DIGBY PARTON

MONDAY, MAY 25, 2015 04:58 AM CDT

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Rebel Wilson lied to us about her age and history because we asked her to

COLIN GORENSTEIN



7 times women shut down sexist reporters

COLIN GORENSTEIN



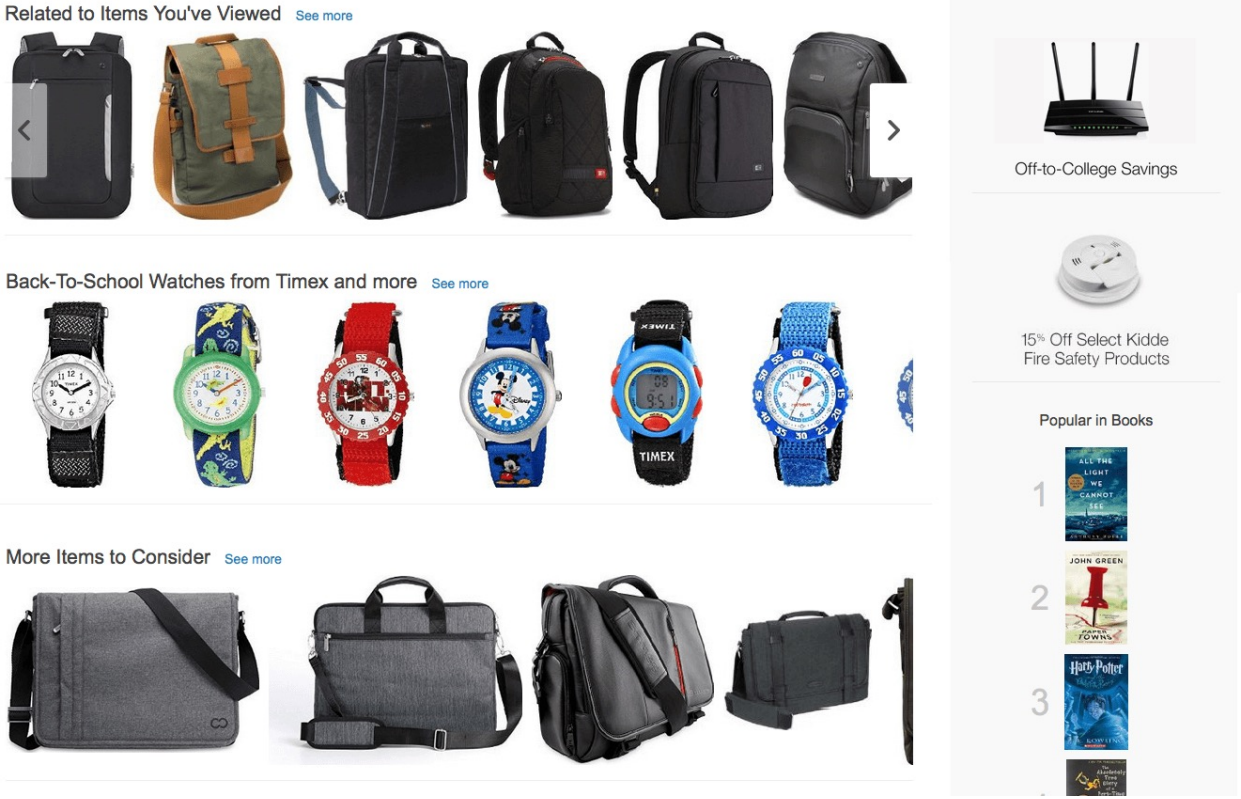
AUDIO: Let's revisit what Michelle Duggar said about LGBT people

Similarity

- When we're designing we can use similarity to convey organization and associations by showing which elements are related, or not related, to one another
- For example, the two sections shown on Salon's site of the previous slide are grouped by their relative sizes
- We clearly perceive two separate groups, with top stories on the left and most read on the right
- Even though both sections provide the same function — displaying articles to read — **grouping by size** assigns more salience to the left grouping, **which allows Salon to guide attention to their top stories**

Similarity

- In contrast, the lack of size differences in this example from Amazon doesn't assign any particular importance to any one section. Despite this, similarity is still achieved through the use of repetition of similar size and shape, causing us to perceive rows of items.



Similarity

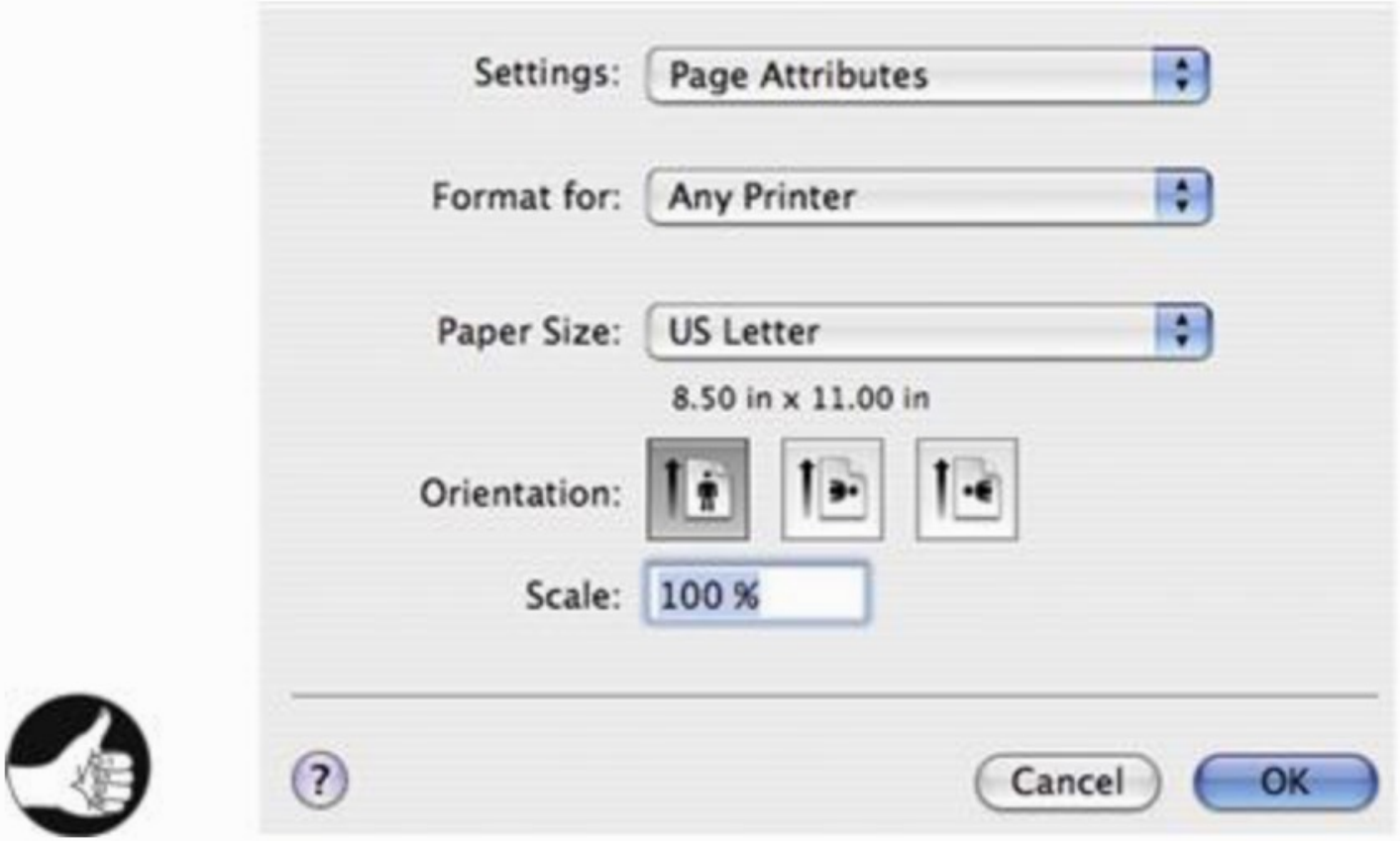


FIGURE 2.6 Mac OS Page Setup dialog box. The Similarity and Proximity principles are used to group the Orientation settings.

Similarity

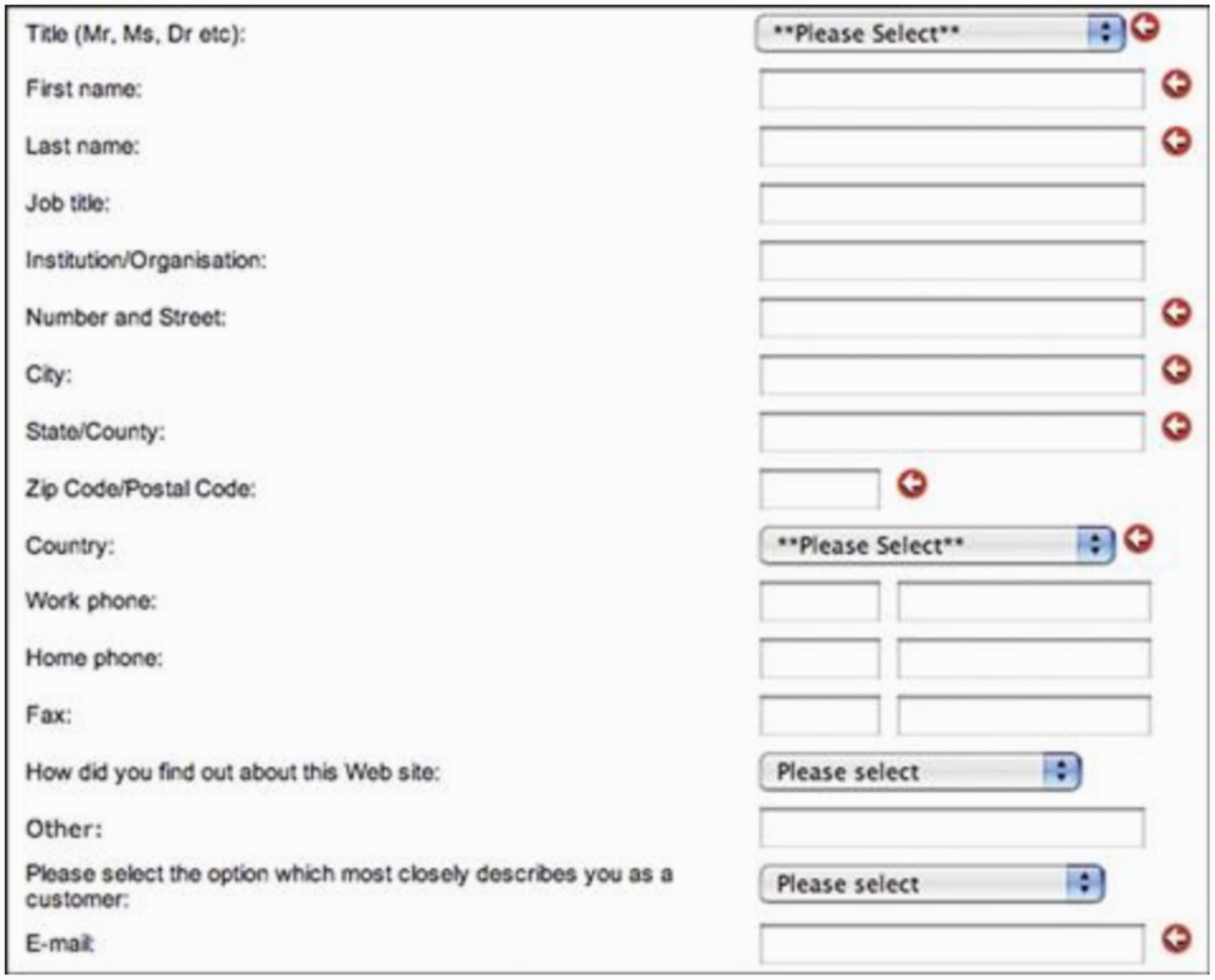
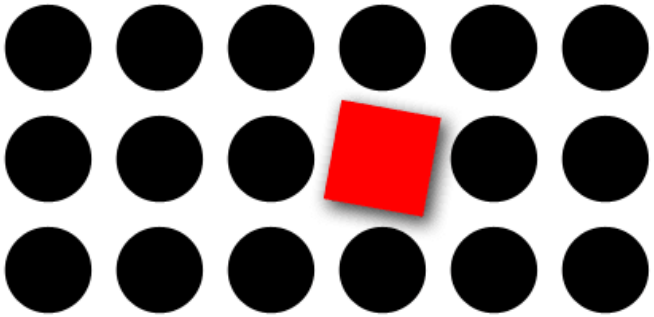




FIGURE 2.7 Similarity makes the text fields appear grouped in this online form at Elsevier.com.

Focal point

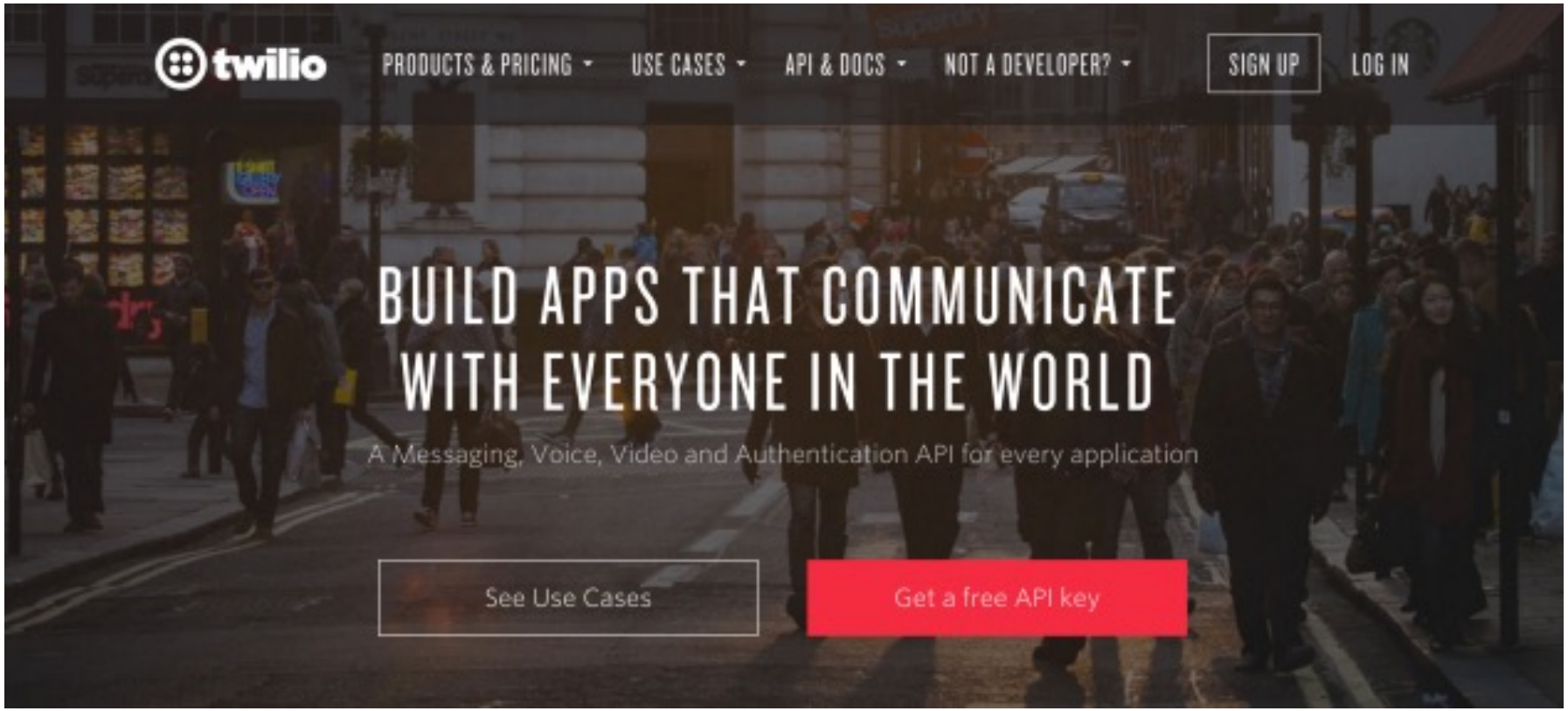
- Distinctive objects create a focal point
- When similarity occurs, an object can be emphasized if it is dissimilar to the others
  - Whatever stands out visually will capture and hold the viewer's attention first





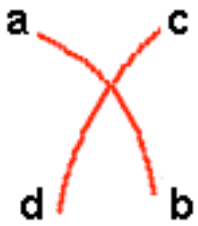
Focal point

- This website uses the focal point principle to draw your eye to their call-to-action button

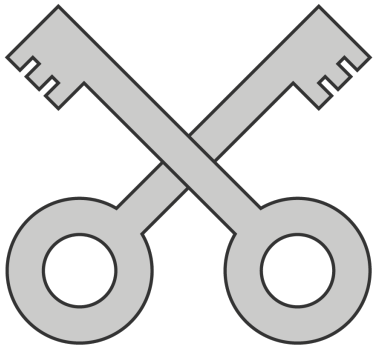


Law of good continuation (or Continuity)

- When there is an intersection between two or more objects, we tend to perceive each object as a single uninterrupted object
- We have a tendency to group and organize lines or curves that follow an established direction over those defined by sharp and abrupt changes in direction
- We have a tendency to perceive unseen parts of an object as continuing in a predictable way







Law of good continuation

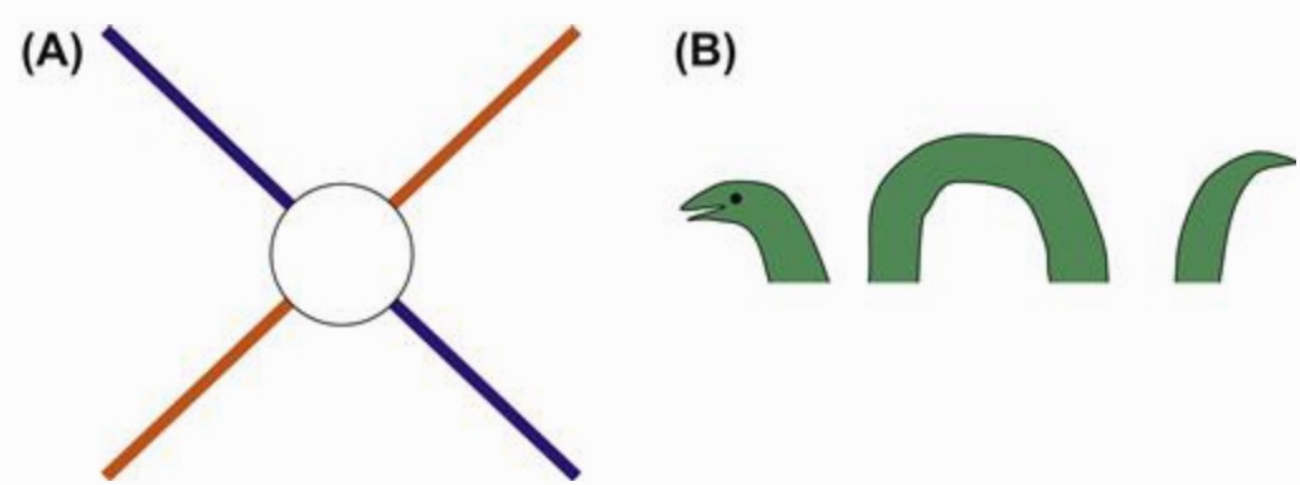


FIGURE 2.8 Continuity: Human vision is biased to see continuous forms, even adding missing data if necessary.



FIGURE 2.9 The IBM company logo uses the Continuity principle to form letters from disconnected patches.

Law of good continuation

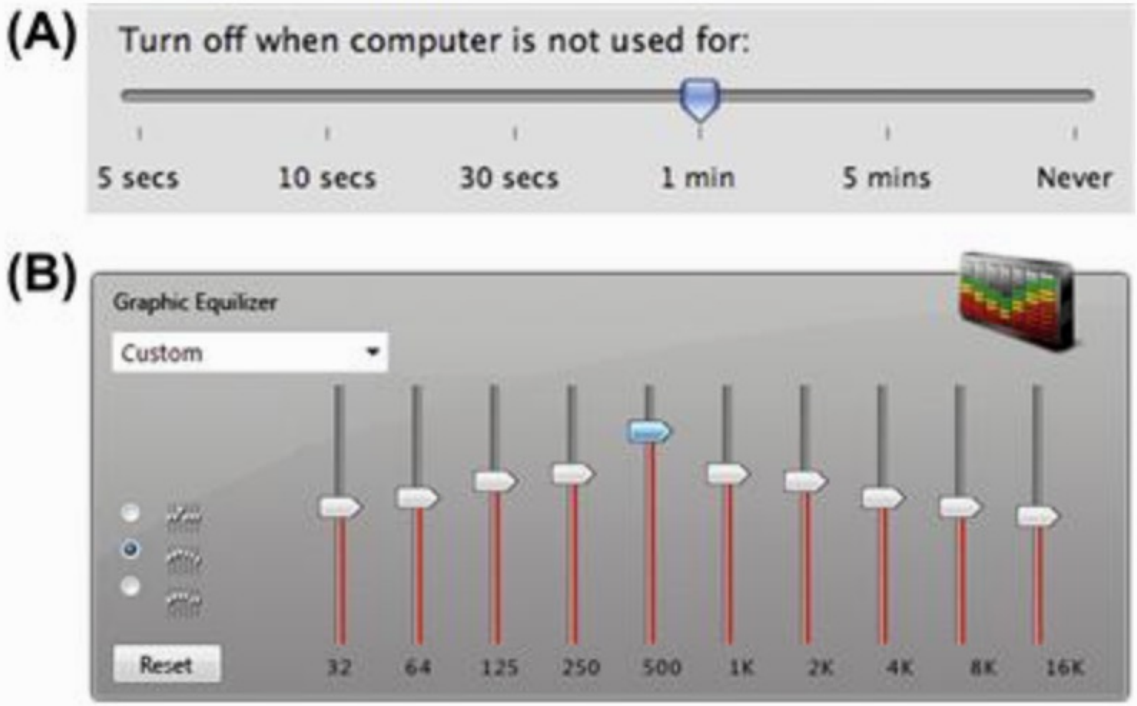
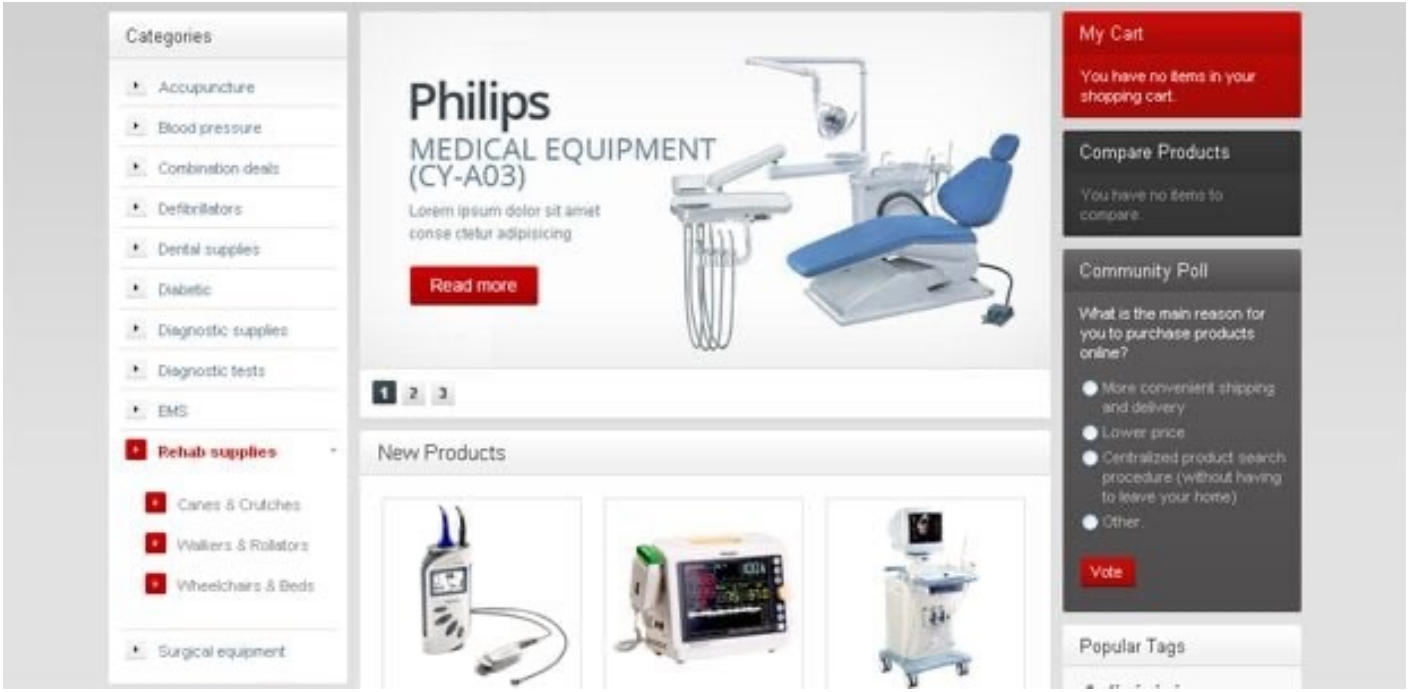


FIGURE 2.10 Continuity: we see a slider as a single slot with a handle somewhere on it, not as two slots separated by a handle: (A) Mac OS and (B) ComponentOne.

Law of good continuation

- The law of continuity says that items placed in a certain row or direction are seen as a group



- Here you can see the main menu items and vertical submenu items that are placed one under another are placed in column as well. The perception of grouped items is strengthened by the law of continuity.

Closure

- Our visual system automatically tries to close open figures so that they are perceived as whole objects rather than separate pieces
- If enough of the shape is indicated, people perceive the whole by filling in the missing information

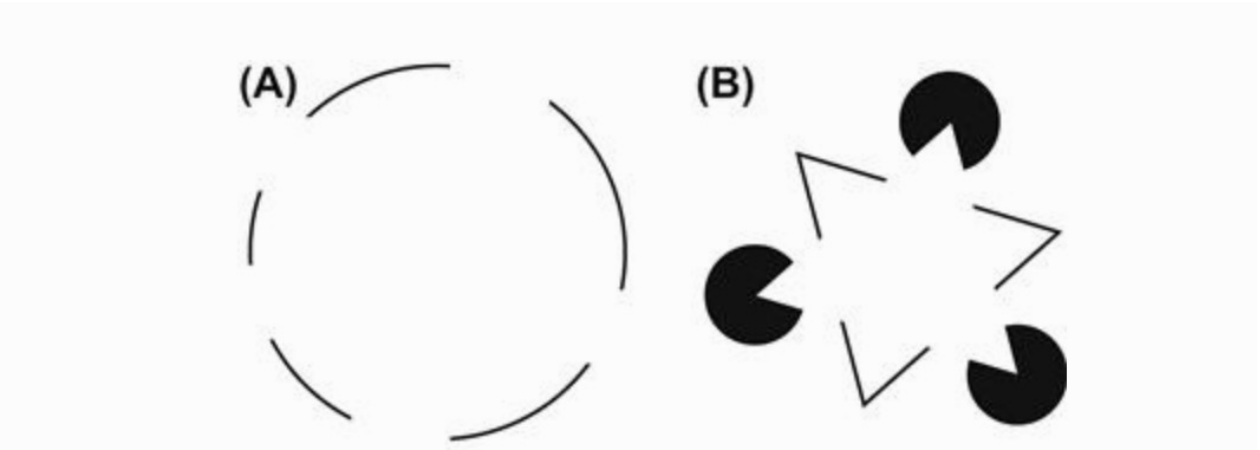
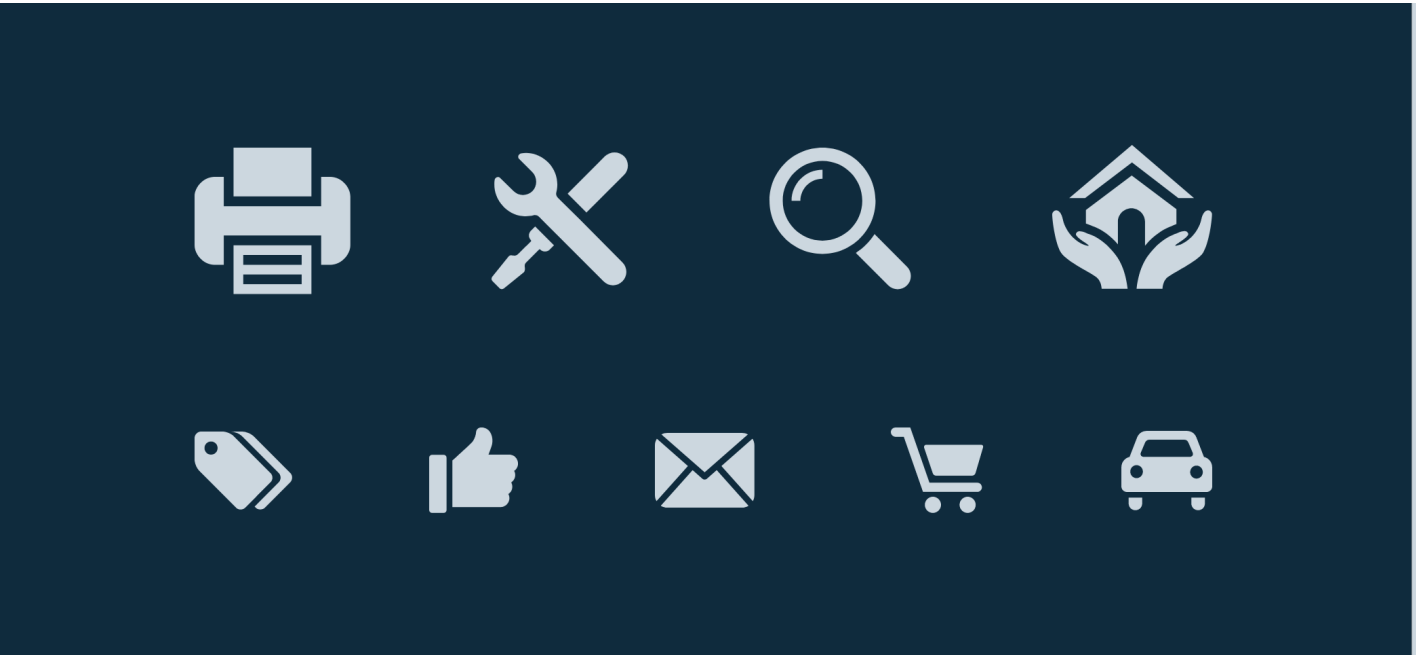


FIGURE 2.11 Closure: Human vision is biased to see whole objects, even when they are incomplete.



Closure

- We can use the Closure principle in Iconography, where simplicity helps with communicating meaning clearly







Closure

- You can't see the text behind the bottle completely, but still know it's 'Kambucha'; except for the first 'K' and the last 'A', other letters are half hidden, which are **intuitively completed and the whole form begins to take shape and consequently, the meaning**



Closure

- Even though the text is not spelled out quite clearly, our minds can easily read the word CULT; the arrangement and alignment of the text made it easy to decipher, even when the words are incomplete



Symmetry

- We tend to parse complex scenes in a way that reduces the complexity
- The data in our visual field usually has more than one possible interpretation, but our vision automatically organizes and interprets the data so as to simplify it and give it symmetry

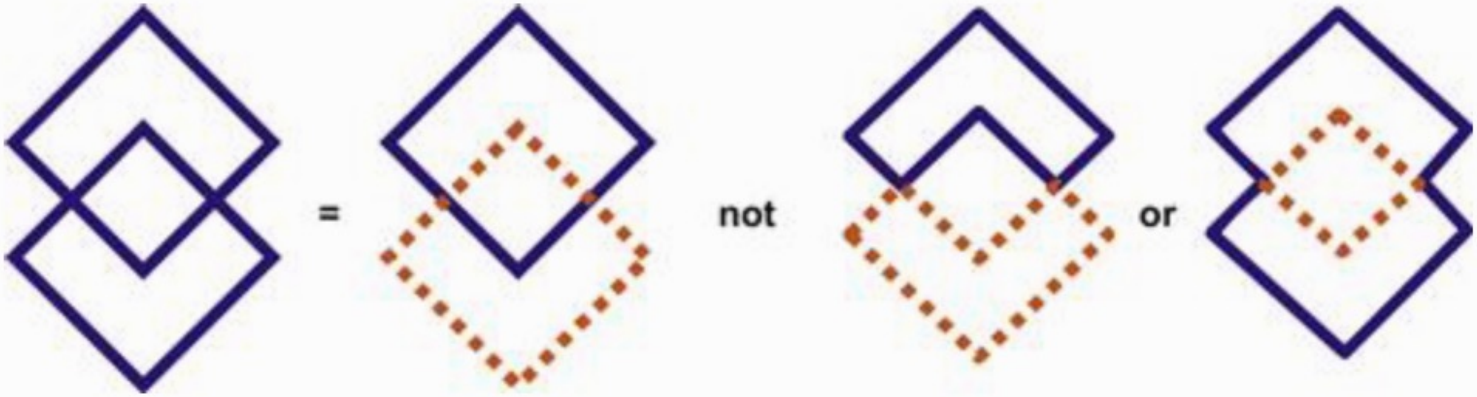
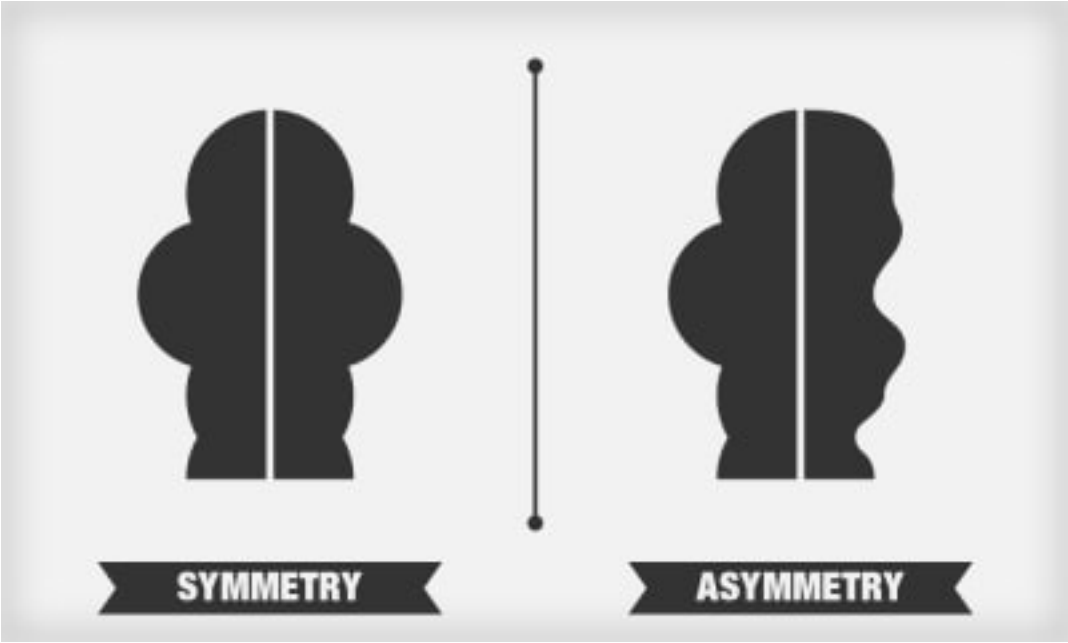


FIGURE 2.13 Symmetry: the human visual system tries to resolve complex scenes into combinations of simple, symmetrical shapes.

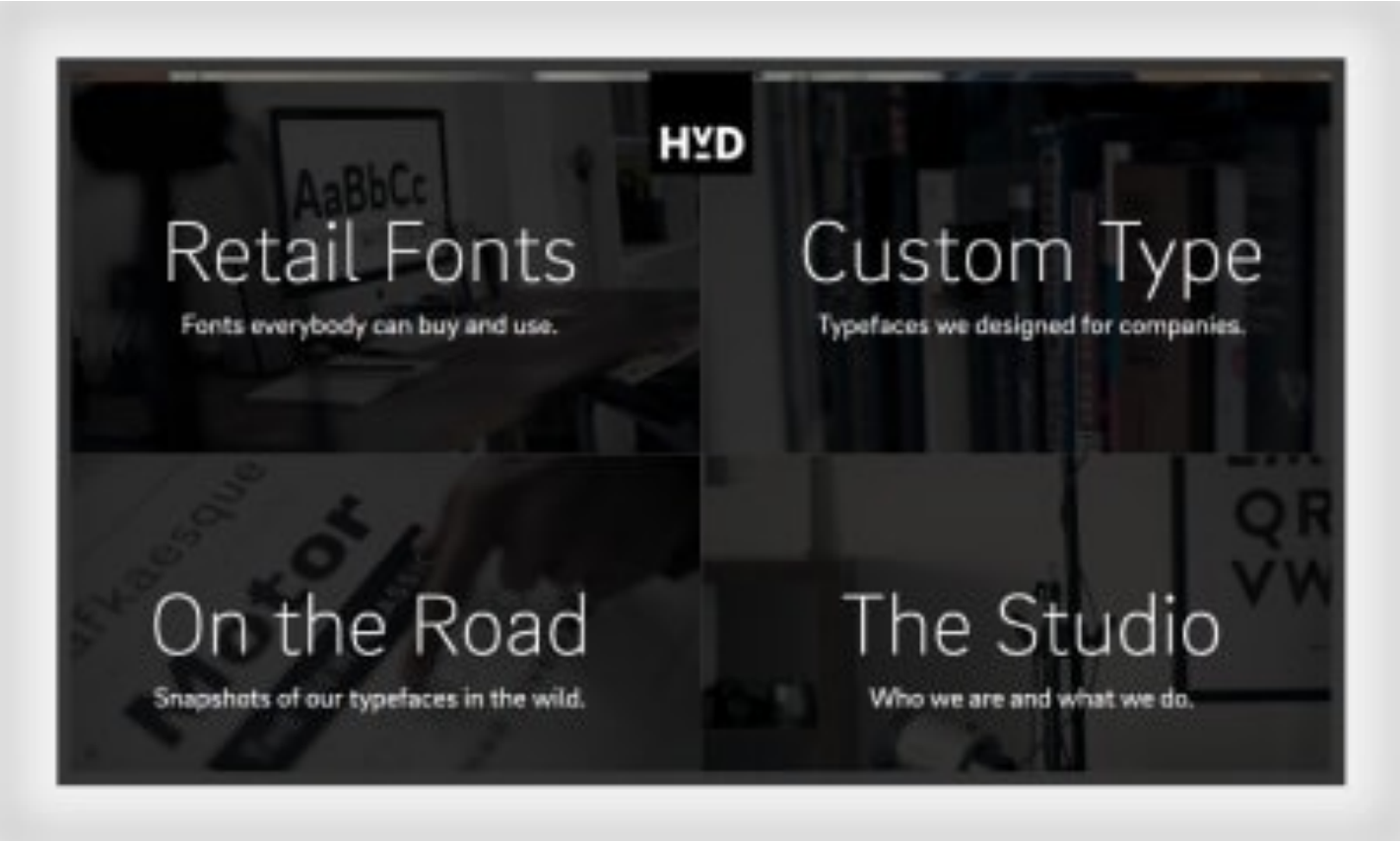
Symmetry

- **The mind tends to find symmetry in objects**, such as images because symmetry creates harmony and allows the viewers to be at comfort while viewing the image



Symmetry

- The page is dissected into four equal halves with fonts of similar size and color; the background in each of the four sections runs along the same theme with a grayscale effect



Common Fate

- Objects that move together are perceived as grouped

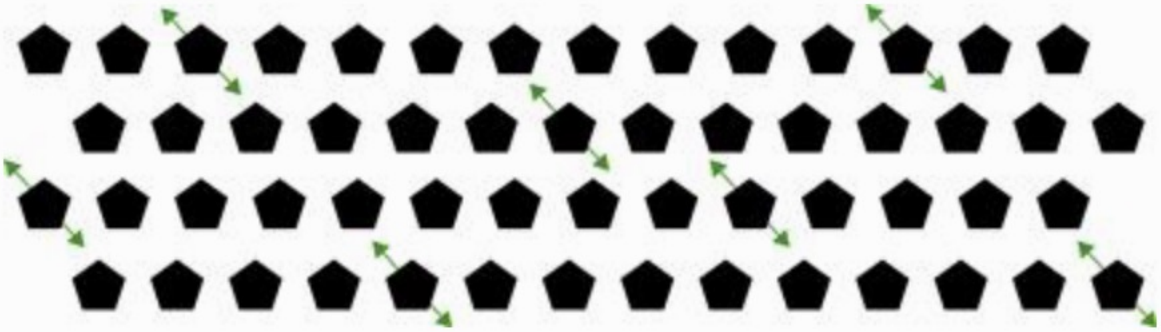
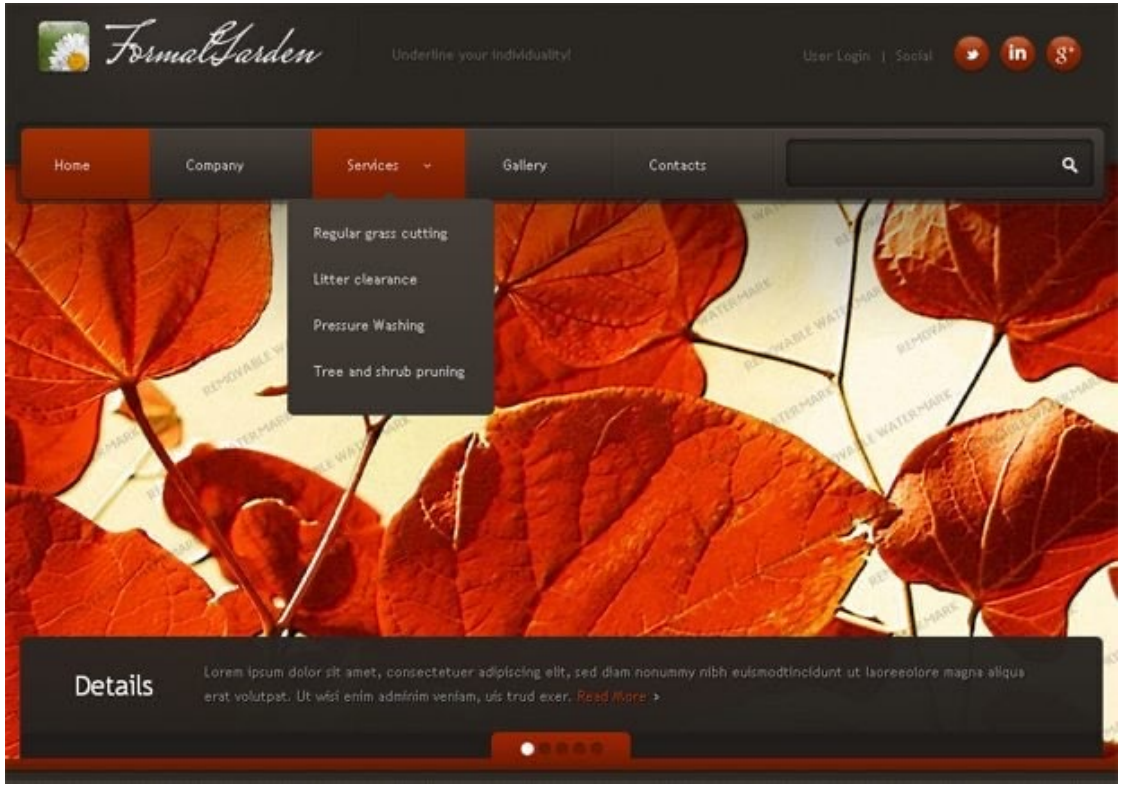


FIGURE 2.20 Common Fate: items appear grouped or related if they move together.

Common Fate

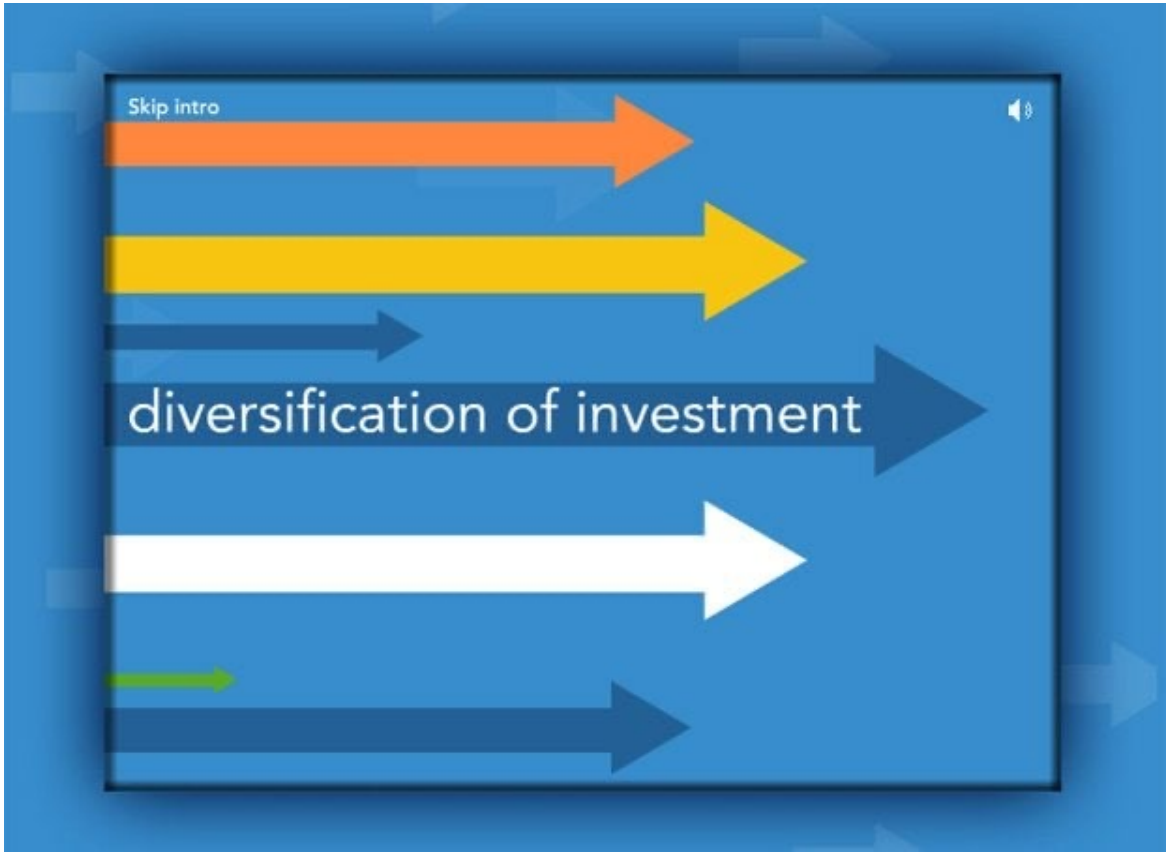
- Common fate principle presumes that as soon as the movement is noted, the relations between objects start to develop



- E.g., drop-down menu: the way how sub-menu items move into visibility shows us that the grouped links are related (within each group)

Common Fate

- Objects that move together are perceived as grouped



Combination of Gestalt principles

- In real-world visual scenes, the Gestalt principles work in concert not in isolation

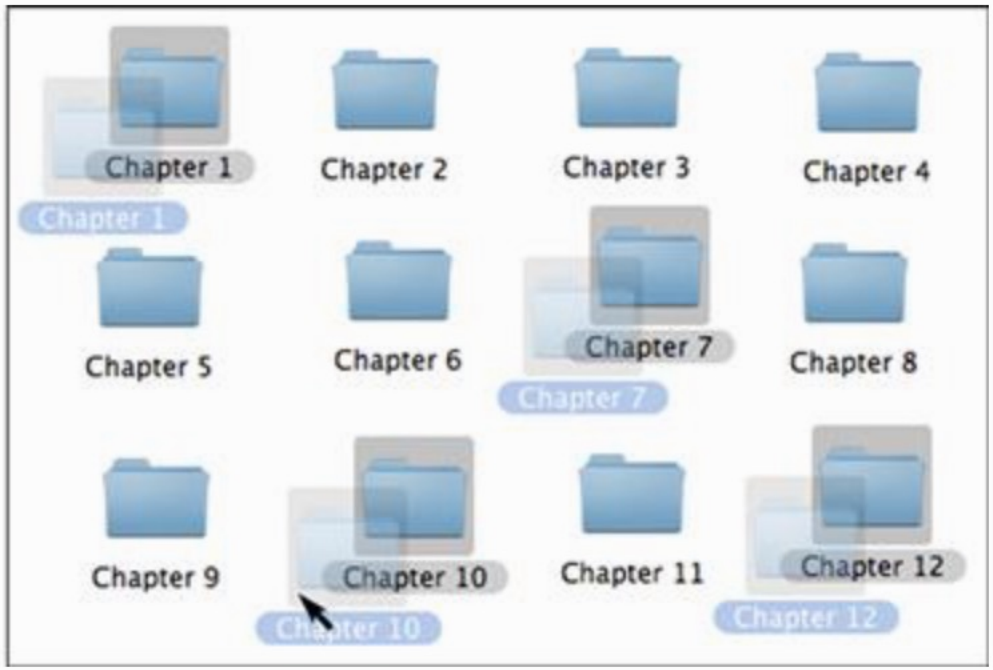
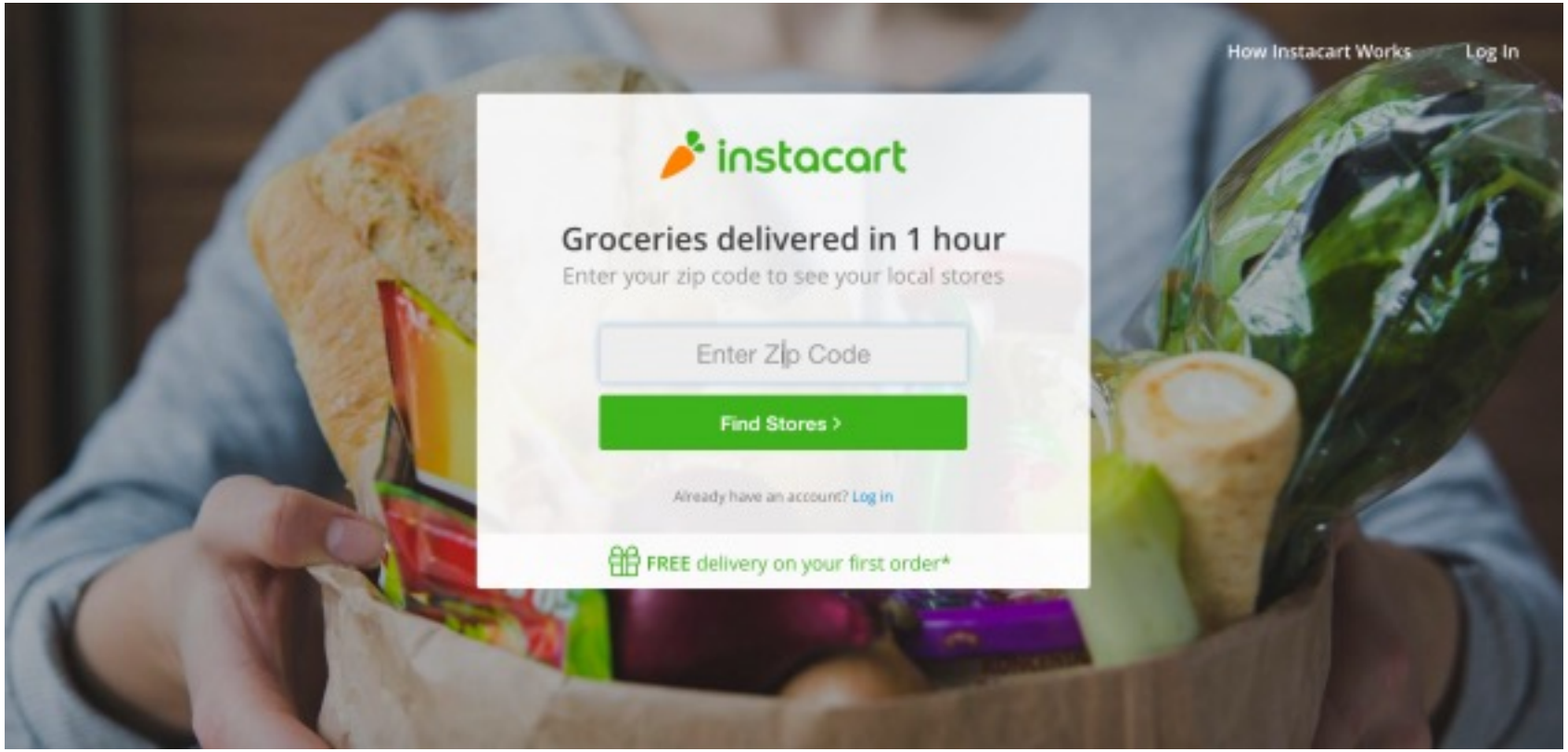


FIGURE 2.23 Similarity and Common Fate: when users drag folders that they have selected, common highlighting and motion make the selected folders appear grouped.

Combination of Gestalt principles

- This website combines the focal point principle with the figure-ground principle to draw your eye to the white foreground, and then to the green "find stores" button



Combination of Gestalt principles

- With all these Gestalt principles operating at once, **unintended visual relationships may be implied by a design**
  - A recommended practice, after designing a display, is to view it with each of the Gestalt principles to see if the design suggests any relationships between elements that you do **not** intend

The importance of providing visual structure

- The Gestalt principles show how our visual system is optimized to perceive structure
- Interactive systems that display information in structured ways allow people to **extract the desired information more easily and quickly**
- When people are navigating through software or Web sites, they don't scrutinize screens carefully and read every word. They scan quickly for relevant information
- **The more structured and concise the presentation of information, the more quickly and easily people can scan and comprehend it**

The importance of providing visual structure

- Example



FIGURE 3.1 Structured presentation of airline reservation information is easier to scan and understand.

Pennsylvania
Bedford Motel/Hotel: Crinaline Courts
(814) 623-9511 S: $18 D: $20
Bedford Motel/Hotel: Holiday Inn
(814) 623-9006 S: $29 D: $36
Bedford Motel/Hotel: Midway
(814) 623-8107 S: $21 D: $26
Bedford Motel/Hotel: Penn Manor
(814) 623-8177 S: $19 D: $25
Bedford Motel/Hotel: Quality Inn
(814) 623-5189 S: $23 D: $28
Bedford Motel/Hotel: Terrace
(814) 623-5111 S: $22 D: $24
Bradley Motel/Hotel: De Soto
(814) 362-3567 S: $20 D: $24
Bradley Motel/Hotel: Holiday House
(814) 362-4511 S: $22 D: $25
Bradley Motel/Hotel: Holiday Inn
(814) 362-4501 S: $32 D: $40
Breezewood Motel/Hotel: Best Western Plaza
(814) 735-4352 S: $20 D: $27
Breezewood Motel/Hotel: Motel 70
(814) 735-4385 S: $16 D: $18

South Carolina

City	Motel/Hotel	Area code	Phone	Rates Single	Rates Double
Charleston	Best Western	803	747-0961	$26	$30
Charleston	Days Inn	803	881-1000	$18	$24
Charleston	Holiday Inn N	803	744-1621	$36	$46
Charleston	Holiday Inn SW	803	556-7100	$33	$47
Charleston	Howard Johnsons	803	524-4148	$31	$36
Charleston	Ramada Inn	803	774-8281	$33	$40
Charleston	Sheraton Inn	803	744-2401	$34	$42
Columbia	Best Western	803	796-9400	$29	$34
Columbia	Carolina Inn	803	799-8200	$42	$48
Columbia	Days Inn	803	736-0000	$23	$27
Columbia	Holiday Inn NW	803	794-9440	$32	$39
Columbia	Howard Johnsons	803	772-7200	$25	$27
Columbia	Quality Inn	803	772-0270	$34	$41
Columbia	Ramada Inn	803	796-2700	$36	$44
Columbia	Vagabond Inn	803	796-6240	$27	$30





- In the left screen the information is bunched up together, making it hard to search
- In the right screen the characters are grouped into vertical categories of information making it easier to search



Renewals, Duplicates, and Information Changes for Driver Licenses and/or ID Cards

- How to renew your driver license in person
- How to renew your driver license by mail
- How to renew your driver license by Internet
- How to renew your instruction permit
- How to apply for a duplicate driver license or identification (ID) card
- How to change your name on your driver license and/or identification (ID) card
- How to notify DMV of my change of address
- How to register for the organ donor gift of life program

FIGURE 3.2 Contents page at the California Department of Motor Vehicles (DMV) Web site buries the important information in repetitive prose.



Licenses & ID Cards:	**Renewals, Duplicates, Changes**		
• Renew license:	in person	by mail	by Internet
• Renew:	instruction permit		
• Apply for duplicate:	license	ID card	
• Change of:	name	address	
• Register as:	organ donor		

FIGURE 3.3 California DMV Web site Contents page with repetition eliminated and better visual structure.

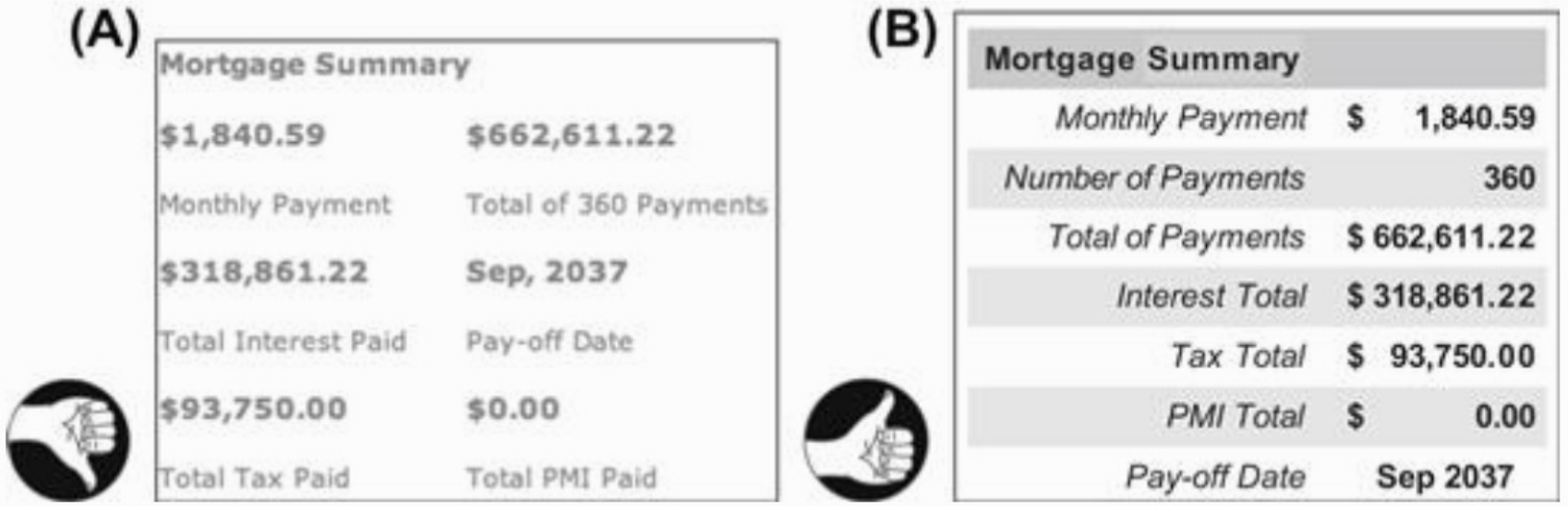


FIGURE 3.5 (A) Mortgage summary presented by a software mortgage calculator; (B) an improved design.

Structure enhances people's ability to scan long numbers

Easy:	(415) 123-4567
Hard:	4151234567
Easy:	1234 5678 9012 3456
Hard:	1234567890123456

FIGURE 3.6 Telephone and credit card numbers are easier to scan and understand when segmented.

Structure enhances people's ability to scan long numbers

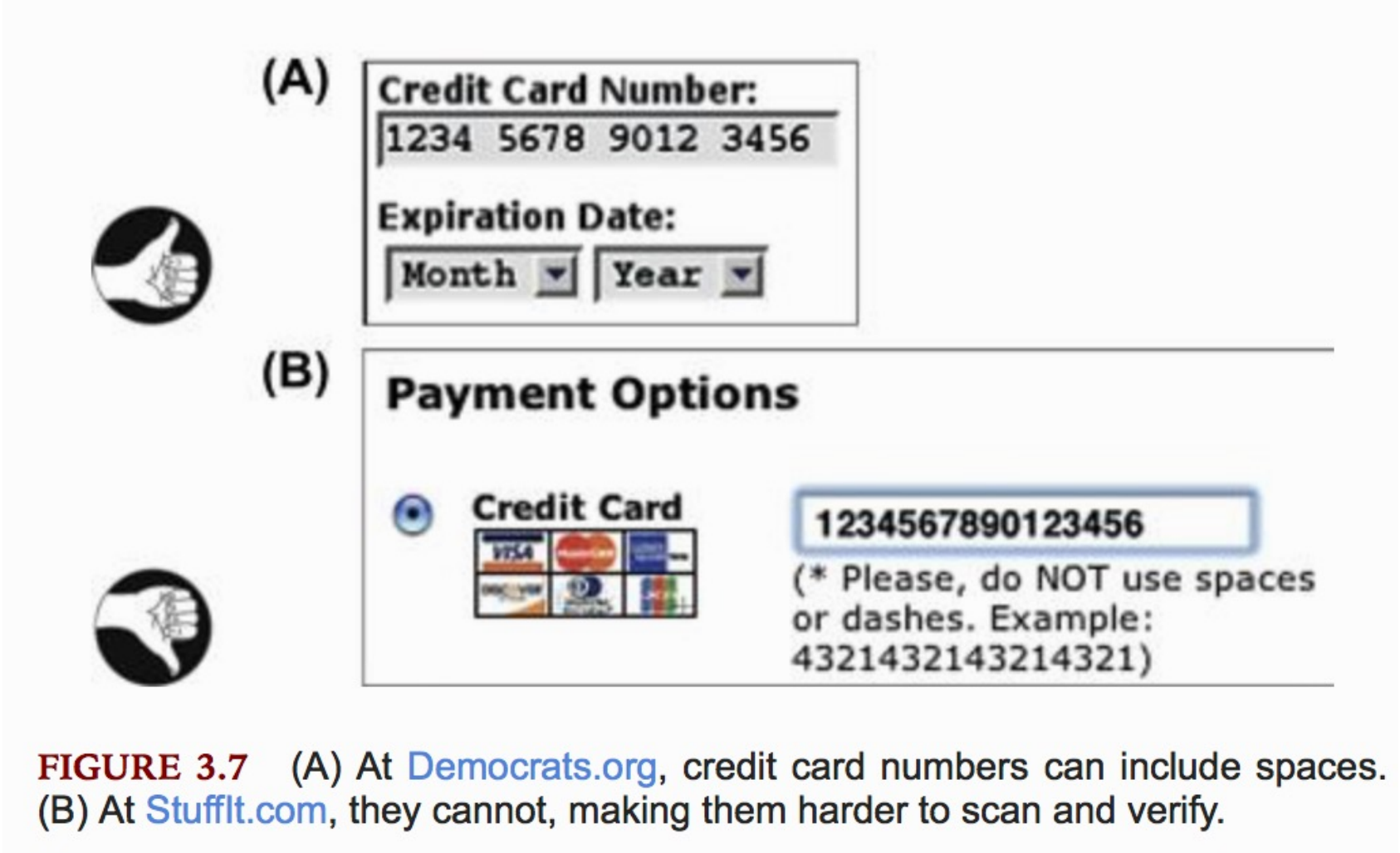


FIGURE 3.7 (A) At Democrats.org, credit card numbers can include spaces. (B) At StuffIt.com, they cannot, making them harder to scan and verify.

Dates

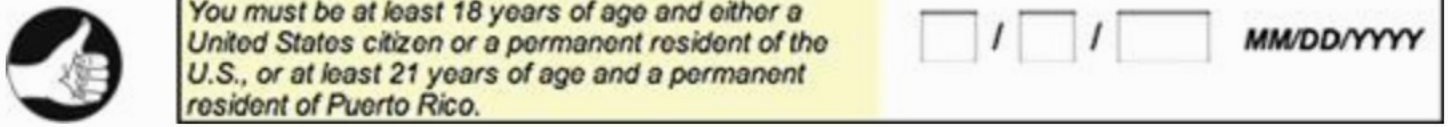


FIGURE 3.8 At BankOfAmerica.com, segmented data fields provide useful structure.

- Good design because it provides structure
- A better design is to use controls that display a value of specific type rather than free text (which could lead to errors when typing)



FIGURE 3.9 At NWA.com, dates are displayed and entered using a control that is specifically designed for dates.

Visual hierarchy lets people focus on the relevant information

- An important goal in presenting information is to provide a **visual hierarchy**, i.e., an arrangement of the information that:
  - Breaks the information into distinct sections, and breaks large sections into subsections
  - Labels each section and subsection prominently to clearly identify their content
  - Presents higher-level sections more strongly than lower-level ones
- This allows people, when scanning information, to **separate what is relevant to their goals** from what is irrelevant **instantly**, and to focus on the relevant information





(A)

Create a Clear Visual Hierarchy

Organize and prioritize the contents of a page by using size, prominence, and content relationships. Let's look at these relationships more closely. The more important a headline is, the larger its font size should be. Big bold headlines help to grab the user's attention as they scan the Web page. The more important the headline or content, the higher up the page it should be placed. The most important or popular content should always be positioned prominently near the top of the page, so users can view it without having to scroll too far. Group similar content types by displaying the content in a similar visual style, or in a clearly defined area.

(B)

Create a Clear Visual Hierarchy

Organize and prioritize the contents of a page by using size, prominence, and content relationships.

Let's look at these relationships more closely:

- **Size.** The more important a headline is, the larger its font size should be. Big bold headlines help to grab the user's attention as they scan the Web page.
- **Prominence.** The more important the headline or content, the higher up the page it should be placed. The most important or popular content should always be positioned prominently near the top of the page, so users can view it without having to scroll too far.
- **Content Relationships.** Group similar content types by displaying the content in a similar visual style, or in a clearly defined area.

FIGURE 3.11 Find the advice about prominence in each of these displays. Prose text format (A) makes people read everything. Visual hierarchy (B) lets people ignore information irrelevant to their goals.



FIGURE 3.12 Visual hierarchy in interactive control panels and forms lets users find settings quickly: (A) Band-in-a-Box (bad) and (B) GarageBand (good).

Lesson for the course “Human-Computer Interaction”

Axure RP (Part 4)

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UNIVERSITY OF TRENTO - Italy

Tutorial

- https://docs.axure.com/tutorials/mobile-patterns/swiping-slideshow/
- https://docs.axure.com/tutorials/mobile-patterns/toggle-switch/

Homework

- **Analyze a website and an app, identifying**
  - Which gestalt principles are applied
  - The presence of structured or non-structured information
  - Reflect on the motivation of the design choices that were made

- Read the sections "Widgets" and "Pages" on the Axure documentation page: https://docs.axure.com

Bibliography

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- https://blog.teamtreehouse.com/gestalt-principles-designers-applying-visual-psychology-modern-day-design
- https://www.smashingmagazine.com/2019/04/spaces-web-design-gestalt-principles